Prabhavit Gupta

Forward Deployed Engineer | Agentic Systems, Open LLM Orchestration & Enterprise AI Solutions

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Executive Summary

- **Forward Deployed Engineering & Applied AI Delivery:** Forward Deployed AI Engineer with 3 years commercial software experience partnering with enterprise customers to translate cutting-edge open LLMs into high-impact, production-grade agentic applications and automated workflows.
- **Agentic Systems & LLM Orchestration:** Deep hands-on expertise building production agent architectures, multi-step tool execution pipelines, Model Context Protocol (MCP) integrations, structured outputs, and hybrid retrieval systems (RAG, pgvector).
- **Enterprise Deployment, Evaluation & Customer Agency:** Experienced embedding directly with customer technical teams to architect solutions, support VPC/cloud deployments, conduct rigorous evals (DeepEval/RAGAS), and establish repeatable deployment playbooks in fast-paced startup environments.

Technical Skills & Core Competencies

- **Agentic Architectures & AI Stack:** Agent Orchestration, Tool Calling, Model Context Protocol (MCP), Open Foundation Models, RAG Pipelines, Vector Databases (pgvector, ChromaDB), Prompt Engineering, Fine-Tuning workflows.
- **Core Software Engineering:** Python (Async, Typing, PyTest), TypeScript, Node.js, Next.js, REST/GraphQL APIs, Structured Data Validation (Pydantic, Zod), Microservices.
- **Enterprise DevOps & Cloud:** Docker, CI/CD (GitHub Actions), Cloud Environments (AWS, GCP, VPC/Hybrid), API Gateways, Secure Webhooks, Sandboxed Tool Execution.
- **AI Observability & Evaluations:** Distributed Tracing (Langfuse), DeepEval, RAGAS, Evals Pipelines, Token/Cost Monitoring, Hallucination Prevention.
- **Forward Deployed Delivery:** Customer Technical Discovery, Solution Architecture, Enterprise Deployment Playbooks, Technical Prototyping, Stakeholder Communication.

Professional Experience

Forward Deployed AI Engineer & Solutions Consultant • *Independent / AI Product Engineering Consultancy*
2024 – Present • Venice, IT (Remote / Hybrid)

- Partnered with enterprise leadership and technical teams from discovery through production launch to architect and deploy customer-specific agentic systems and automated AI pipelines.
- Engineered agent execution runtimes integrating LLMs with external enterprise systems and internal APIs via Model Context Protocol (MCP) and custom tool-calling interfaces.
- Architected enterprise RAG and semantic extraction systems using PostgreSQL (pgvector) and hybrid search, optimizing context retrieval precision and reducing inference latency.
- Established automated evaluation harnesses using DeepEval and distributed observability via Langfuse to track task completion accuracy, latency profiles, and cost attribution.
- Authored technical implementation playbooks, client integration documentation, and architectural designs delivered directly to executive and engineering stakeholders.

Full-Stack AI Software Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Shipped production web services and automation pipelines in Python and TypeScript, integrating LLM APIs and modern data orchestration workflows.
- Constructed automated scraping and ingestion pipelines handling thousands of unstructured documents with AI extraction, validation, and programmatic sync to enterprise CRMs.
- Implemented CI/CD pipelines with automated test suites in Docker, guaranteeing zero-downtime releases and high software reliability.
- Collaborated with product and business stakeholders to turn real-world operational challenges into scalable, user-centric software features.

Key Engineering Projects

- **Enterprise Agentic Tool Gateway (MCP + Python + Docker):** Built a sandboxed agent runtime connecting frontier and open LLMs to enterprise databases and internal developer tools with automated policy guards and full audit tracing.
- **AI Evaluation & Tracing Harness (Langfuse + DeepEval + FastAPI):** Implemented an evaluation pipeline running synthetic test suites and live trace benchmarking across multi-agent workflows to prevent regression.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy & Economics** – Ca' Foscari University of Venice (2020 – 2024)
  Rigorous training in formal logic, epistemology, decision theory, and analytical problem-solving.
- **Languages:** English (C1 Full Professional / Fluent), Italian (Working proficiency), Hindi (Native).
- **Legal Status:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy) – Unrestricted work authorization across the EU; eligible for relocation and visa sponsorship.